

NUU Cloud Plan (Gemini)

NUU Cloud: Project Specification, Documentation, and Architecture

This document outlines the specification, documentation, and technical architecture for NUU Cloud, the foundational data storage and synchronization service for the NUU Cognition ecosystem.

1. Project Specification

1.1 Project Overview

NUU Cloud is the central data layer and backend infrastructure for the entire NUU ecosystem, supporting applications such as NUU Surf, NUU Capture, and NUU Echo, as well as the internal organizational knowledge base (NUU Core). It is designed as a robust, "bread-and-butter" service responsible for storing, managing, and retrieving the ecosystem's core data: Markdown files and associated media.

1.2 Purpose and Vision

The vision for NUU Cloud is to provide a unified infrastructure for managing knowledge and data. It is designed to support a local-first architecture while offering the convenience, accessibility, and collaborative features of a cloud service. It aims to be the centralized, searchable repository powering both individual cognition (Cognition OS) and collaborative workspaces (NUU Collective).

1.3 Core Philosophy

1.3.1 Markdown as the Universal Format

The cornerstone of the NUU ecosystem is the exclusive use of Markdown (`.md`) as the base format for all textual data. NUU Cloud is optimized for storing, parsing, and serving Markdown files. This decision ensures:

- **Data Ownership:** Users retain control over their data in a universally readable, portable format.
- **Interoperability:** Simplifies data transfer and interaction between different NUU applications.
- **AI Compatibility:** Allows for seamless integration with AI/LLMs, which natively process Markdown, facilitating context ingestion.

1.3.2 Local-First Approach

The ultimate goal is a local-first architecture where users own their data in local folders (Cognition OS structure). While the initial product launches (like NUU Surf) will be cloud-based for accessibility and speed of development, NUU Cloud is architected to eventually serve as the synchronization layer between local instances and the cloud repository. The architecture mandates standardized business logic that can operate identically whether hosted locally or in the cloud.

1.4 Functional Requirements (MVP)

The Minimum Viable Product (MVP) focuses on supporting the NUU Surf beta launch (Target: Mid-August 2025).

- **Data Storage and Management:** Persistent storage for Markdown files and associated image files.
- **Authentication (Auth):** Handle user registration, login, and session management. Integration with Clerk is prioritized due to its generous free tier (up to 10,000 MAU).
- **Authorization:** Implement controls to ensure users can only access their own files.
- **File Operations (CRUD):** Expose an API endpoint for Create, Read, Update, and Delete operations on Markdown and image files.
- **Image Handling:** Ensure that when a Markdown file is retrieved, referenced images are also accessible via secure URLs.

1.5 Future Functional Requirements

- **Syncing Protocol:** Develop a protocol to synchronize the cloud storage layer with local Cognition OS folders.
- **Version Control:** Implement version control for Markdown files, allowing users to track changes and roll back ("live version rollback").
- **Audio Storage:** Support for audio files to enable NUU Echo.
- **Advanced Sharing:** Implement group access and advanced sharing permissions for NUU Collective.

1.6 Non-Functional Requirements

- **Scalability:** The infrastructure must scale to accommodate a growing user base and data storage needs.
 - **Reliability and Robustness:** As the foundational layer, high availability and fault tolerance are required. Data integrity is paramount.
 - **Security:** User data must be secured in transit and at rest, following industry best practices.
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2. Documentation

2.1 The Three-Layer Architecture

The NUU ecosystem operates on a three-layer architecture model. NUU Cloud is central to the first layer and facilitates the second.

1. **State/Storage Layer (Data):** The foundational layer where the system's state resides (Markdown and media). This exists in two forms:
 - **Local:** User-owned folders on their device (Cognition OS).
 - **Cloud (NUU Cloud):** Cloud-based storage mirroring the local structure.
2. **Logic Layer (Business/Parsing):** The business logic, parsing (e.g., NUU Surf backend), and data processing protocols. This layer interacts with the Storage Layer via defined gateways (e.g., NUU Cloud APIs). The goal is for this logic to be standardized and environment-agnostic (capable of running locally or in the cloud).
3. **UI Layer (Presentation):** The front-end applications (e.g., NUU Surf UI, Obsidian) through which users interact with the system.

2.2 Ecosystem Integration

NUU Cloud is the central infrastructure interacting with various specialized services:

2.2.1 NUU Capture

NUU Capture is the gateway service responsible for converting external information (PDFs, websites, audio transcripts, images) into the standardized NUU Markdown format. Once processed (e.g., via OCR and LLM refinement), NUU Capture uploads the resulting Markdown and associated media to NUU Cloud for persistent storage.

2.2.2 NUU Surf

NUU Surf relies on NUU Cloud for data access. The backend logic for NUU Surf (the hierarchical summarization process) reads the base Markdown from NUU Cloud and potentially writes the generated summaries ("Surf Markdown") back to NUU Cloud for caching. The NUU Surf UI fetches this data for display.

2.2.3 NUU Core (The Conceptual Data Layer)

NUU Cloud hosts "NUU Core" (the "cast"), the organization's internal knowledge vault. This is a conceptual, descriptive layer of the organization stored in Markdown. It defines protocols, documentation, and dependencies (e.g., linking files via `[[filename]]` syntax). AI agents can ingest this context directly from NUU Cloud to understand the system architecture and business logic.

3. Technical Architecture

3.1 High-Level Architecture (MVP)

The architecture utilizes a centralized API service to manage interactions between NUU applications and the underlying data storage, utilizing managed services for rapid development.

Code snippet

```
graph TD
    subgraph UI_Layer [UI Layer]
        A[NUU Surf UI]
        B[NUU Capture UI]
    end

    subgraph Logic_Layer [Logic Layer]
        E["NUU Surf Backend<br/>(Summarization/Parsing)"]
        F["NUU Capture Service<br/>(OCR/Conversion)"]
    end

    subgraph State_Layer [State Layer (NUU Cloud)]
        G["NUU Cloud API (Python)"]
        H["(Database - Supabase/Postgres)"]
        I["(Media Storage - Supabase Storage)"]
        J["{Authentication - Clerk}"]
    end

    A -- Requests --> E
    B -- Uploads Raw --> F

    E -- Read/Write (Processed Data) --> G
    F -- Write (Markdown/Media) --> G

    G -- CRUD Operations --> H
    G -- Store/Retrieve --> I
    G -- Authenticate/Verify --> J
    A -- Authenticate --> J
```

3.2 Technology Stack (MVP)

The initial MVP implementation will utilize the following stack for rapid development and scalability:

- **Backend/API:** Python (e.g., FastAPI or Flask).
- **Database:** Supabase (PostgreSQL).
- **Authentication:** Clerk (Prioritized for generous free tier).

- **Media Storage:** Supabase Storage (Object storage for images).
- **Hosting:** TBD (A suitable platform for hosting the Python service).

3.3 Data Model (MVP Schema)

The initial database schema in Supabase will be simplified. For the MVP, Markdown content will be stored directly in the database as a `TEXT` field rather than managing discrete files.

3.3.1 Users

Managed primarily by Clerk. A local table may link Auth IDs to internal permissions.

Column	Type	Description
user_id	UUID (PK)	Unique internal identifier.
auth_provider_id	Varchar (Unique)	ID from the external Auth provider (e.g., Clerk).
email	Varchar	User's email address.

3.3.2 Documents (Markdown)

The core storage mechanism.

Column	Type	Description
document_id	UUID (PK)	Unique identifier for the document.
user_id	UUID (FK)	Owner of the document.
title	Varchar	Title or name of the document.
markdown_content	TEXT	The raw Markdown content.
created_at	Timestamp	Timestamp of upload.
updated_at	Timestamp	Timestamp of last modification.

3.3.3 Media Assets

Stores images and other media referenced in the documents.

Column	Type	Description
asset_id	UUID (PK)	Unique identifier for the asset.
user_id	UUID (FK)	Owner of the asset.
document_id	UUID (FK)	Associated Markdown document (Optional).
storage_url	Varchar	Link/Path to the asset in Object Storage.
file_type	Varchar	Type of asset (e.g., 'image/png').

3.4 API Specification (Key Endpoints)

NUU Cloud will expose a RESTful API for internal services.

- `POST /api/v1/documents` : Upload a new Markdown document.
 - `GET /api/v1/documents/{document_id}` : Retrieve a specific document and its metadata.
 - `PUT /api/v1/documents/{document_id}` : Update a document.
 - `GET /api/v1/documents` : List all documents for the authenticated user.
 - `POST /api/v1/media` : Upload media files, returning a storage URL.
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4. Roadmap

Phase 1: MVP (Target: Mid-August 2025)

The immediate goal is to develop the infrastructure required to support the NUU Surf beta launch.

- Implement Python backend and Supabase database/storage.
- Integrate Authentication (Clerk).
- Develop core CRUD APIs for Markdown and Images.
- Enable NUU Surf to fetch/store data via NUU Cloud instead of local files/manual scripts.

Phase 2: Local Sync and Versioning

- Develop the local-to-cloud synchronization protocol (Cognition OS Sync).
- Implement basic version control (e.g., snapshots or "live version rollback") for Markdown files.
- Enhance sharing and collaboration features for NUU Collective.

Phase 3: Scalability and Full Ecosystem Support

- Implement advanced indexing and semantic search capabilities.
- Full integration with NUU Echo (including audio storage).
- Transition from MVP tech stack to more scalable infrastructure if required.